Paleomagnetic Results from the Upper Triassic Lavas of Massachusetts

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Abstract. The directions of magnetization at five sites in two lava members of the Newark group of Massachusetts have been measured. The directions have fairly consistent northerly declinations, but the inclinations are highly variable. This variation is shown to be due to the presence of a soft magnetic component of variable magnitude approximately parallel to the present earth's field, the effect of which is removed in alternating magnetic fields of 150 oersteds. The magnetization remaining after the removal of this 'soft' component has a mean direction (010°, +16°) which is consistent with the paleomagnetic pole position at 55°N, 88°E. This agrees well with other Triassic paleomagnetic pole determinations from the United States.

Introduction. In 1956 we collected 16 samples from two lava members in the Newark group of Massachusetts and measured their directions of natural remanent magnetization ($\mathbf{M}_n$). About one-half of these samples were stable magnetically, and from these a provisional mean direction and pole position was published [*Du Bois, Irving, Opdyke, Runcorn, and Banks*, 1957, second entry Table I] for comparison with paleomagnetic results obtained by others for samples taken elsewhere in the United States from Upper Triassic formations. The remainder of the samples contained a substantial viscous component of magnetization imposed by the present earth's magnetic field, and it has now been possible to demagnetize these components by treating the rock in alternating magnetic fields so that the underlying stable magnetization is revealed. This has been done so that all specimens may now be used, and the results, which supersede the earlier values, are given here. The number of sites visited (5) is not entirely satisfactory, but, since we have no opportunity for further sampling, putting the data on record is worth while.

Results. Several separately oriented samples have been taken from two sites in a lava in the Granby tuff [*Emerson*, 1917, map; *Bain*, 1941, p. 266], which becomes the posterior (Hampden) sheet farther south [*Bain*, 1941, p. 268], and three sites in the Holyoke lava member. Both lavas are members of the Meriden formation and part of the Newark group, the Holyoke lava member being older than the lava in the Granby tuff [*Rodgers, Gates, and Rosenfeld*, 1959, p. 16]. Sites and sampling details are given in Table 1. Two disk specimens (2.1-cm diameter, 0.8-cm thickness) have been cut from each sample. The directions and intensity of natural remanent magnetization in all specimens were measured and the directions corrected for geological tilt by simple rotation around the direction of the strike observed in adjacent sediments. These directions, which are plotted in Figure 1, have a predominantly northerly declination, but, with inclinations varying from +82° to −10°, the difference between specimens from the same site often exceeds 20° or 30°. Many of the directions tend towards the present field and it may be immediately suspected that comparatively 'soft' components of magnetization imposed recently by the earth's field are present.

In order to test whether or not such components are present, specimens were subjected to treatment in alternating magnetic fields in the absence of any steady field, using the apparatus and methods described by *Irving, Stott, and Ward* [1961]. Four specimens each from sites 2 and 5 were treated in alternating magnetic fields in steps of 75, 150, 225, and 300 oersteds (peak field H_p), measurements being repeated between each step. After treatment of the speci-

TABLE 1. Summary of Results

Lava Member	Site	Attitude: Azimuth, deg	Attitude: Dip, deg	S	N	$\mathbf{M}_n$: $D°$	$I°$	R	k	$\theta°$	$\mathbf{M}_{(150)}$: $D°$	$I°$	R	k	$\theta°$
Granby	1. Road cutting near Notch Quarry on Route 116 (72°31′25″W, 42°17′40″N)	090	12S	3	6	009	+52	5.89	45	12	014	+37	5.96	125	8
	2. Mountain Park, road cutting on Route 5 near Mt. Tom Reserve (72°36′30″W, 42°16′11″N)	019	12E	3	6	353	+43	5.39	8	29	357	+21	5.98	250	5
Holyoke	3. Farmington, road cutting on Route 6 (72°49′33″W, 41°42′09″N)	355	15E	3	6	355	+16	5.91	55	11	001	+18	5.95	100	8
	4. Ashley Ponds, road cutting on Route 202 (72°39′42″W, 42°10′04″N)	023	13SE	4	8	017	+13	7.89	64	10	020	+03	7.92	88	9
	5. Notch Quarry, quarry on Route 116 (72°31′25″W, 42°18′20″N)	078	19S	3	6	032	+26	4.94	5	36	014	+01	5.78	23	17

Notes: The site details and attitude of sediments adjacent to the collecting sites are given in the first three columns; the attitude is given by the strike azimuth and the dip. S is the number of samples and N the number of specimens. The values in the columns headed $\mathbf{M}_n$ and $\mathbf{M}_{(150)}$ refer to the natural remanent magnetization and the magnetization after treatment in 150 oersteds alternating field; D and I are the declination and inclination of the mean site directions corrected for attitude, R is the resultant, k is a measure of the within site precision [*Fisher* 1953], and θ is the circular standard deviation of the specimen directions (a cone of half angle θ whose axis is the mean direction contains approximately 63 per cent of the observed directions).

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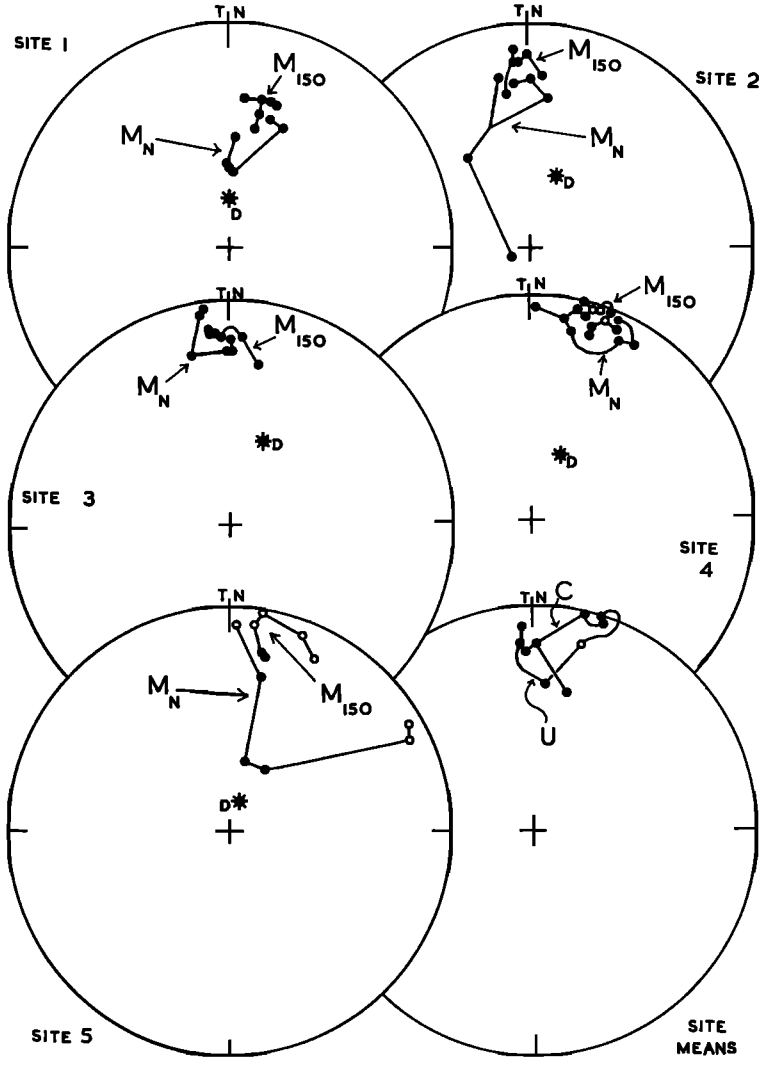


Fig. 1. The observations at each site are plotted on the first five of these equal-area nets, the primitive in each case being the bedding plane measured in adjacent sediments. The initial directions, $\mathbf{M}_n$, are linked together as are those, $\mathbf{M}_{150}$, measured after treatment of the specimens in an alternating magnetic field of 150 oersteds. In all cases the within-site precision improves, this being particularly apparent at sites 2 and 5 where the initial directions often tend strongly towards the present field. At sites 3 and 4 the magnetizations show greater stability. The dipole field direction is labeled D. At bottom right, the site mean directions are plotted before (U) and after (C) correction for the attitude of the beds. In all cases solid dots indicate directions with positive inclination and circles denote negative inclinations.

mens in a field of 150 oersteds the agreement between specimens from the same site was much improved, but in higher fields the directions began to scatter. It is considered that the treatment in 150 oersteds effectively demagnetizes the viscous components, and specimens from all sites have been treated in this way. The agreement in directions both within and between sites is now much improved (see Table 1 and Fig. 1). The inclinations are now all low or moderate.

Many of the specimens, considered individually, were stable, their directions changed little, and their intensities diminished only slowly in increasing alternating magnetic fields; an exam-

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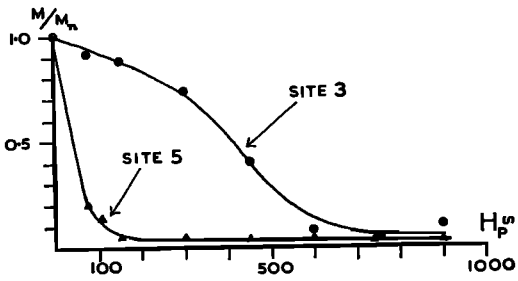


Fig. 2. Effect of alternating magnetic fields on the intensity of magnetization. The ratio of the moment M after treatment to the natural remanence M_n is plotted vertically, and the peak alternating field applied (H_p) along the horizontal axis. The upper curve (dots are experimental points) is for a specimen ($M_n = 1.61 \times 10^{-3}$ emu/cm³) from site 3 (Holyoke lava member), and the lower curve (triangles are experimental points) for a specimen ($M_n = 1.03 \times 10^{-3}$ emu/cm³) from site 5 (Holyoke lava member). These graphs illustrate the variability in stability found within the same lava member.

ple, from site 3 (Holyoke lava member), of the demagnetization curve of such a specimen is given in Figure 2. In this specimen the directions after treatment in a field of 150 oersteds changed by only 5°. In other specimens the magnetization is less stable; their directions, which were initially not far from the present field, change, after treatment in the 150-oersted field, so that they always agree closely with those from stable specimens. The demagnetization curve falls steeply and is of a type characteristic of isothermal remanent magnetization. In these unstable specimens the major part of the natural remanent magnetization is secondary in origin and is aligned approximately along the present field. An example of such a curve from a specimen from site 5 (Holyoke lava member) is given in Figure 2; the directions shift after treatment in 150 oersteds by 46° away from the present field direction.

The over-all mean site direction after correction is (010°, +16°) and the error ($P = 0.05$) in mean (calculated from the over-all precision $k' = 184.5$) is 10°. The between-site precision, calculated by the methods described by *Watson and Irving* [1957], is 41. The mean site directions before and after correction for attitude are plotted in Figure 1. Correction produces a slight improvement in agreement, which indicates stability, but the dips are small and any large effect is not to be expected.

Discussion. The scatter between sites is clearly significant even within the same lava member. Each member may contain several flows, so that this scatter may be due to secular variation of the field during the period in which the magnetization was acquired; it may be noted that the scatter (circular standard deviation 13°) is comparable to the departures from the geocentric dipole field observed nowadays. The simple corrections applied for geological tilt may not always be an adequate restoration and could also be responsible for some of the scatter. For instance, the flexures advocated by *Bain* [1941, p. 265] for the area of Notch Mountain and farther east could have affected site 5 so that the directions should be (014°, −04°); however, the effect is small (cf. Table 1) and would not occur at the other sites.

The paleomagnetic pole positions consistent with the mean direction are 55°N, 88°E, and 55°S, 92°W, which compare very well with those obtained from the Triassic of Connecticut and New Jersey [*Du Bois, Irving, Opdyke, Runcorn, and Banks,* 1957; *Opdyke,* 1961] and also with those obtained from the Triassic of the western United States [*Collinson and Runcorn,* 1960]. This concurrence of paleomagnetic pole results from many different rocks over such a wide area indicates that a dipolar axis with this attitude provides a realistic representation of the Triassic field in the United States. These determinations are, of course, relative to the sampling region only and do not mean that central Siberia was on the paleomagnetic pole at that time; in fact, recent work by *Makarova* [1959] on the Triassic Siberian traps of the Yenesei region indicate that this was not the case.

The inclinations in the lava in the Granby tuff are usually about 10° steeper than those observed in the Holyoke lava member, although it is not possible to test the significance of this difference from the present data. However, it may be possible by more extensive sampling along the outcrops, together with a similar extension of work farther south, to define characteristic direction for each flow and thus to test the correlation, usually assumed by stratigraphers of these lava flows in the upper part of the Newark group of the eastern United States.

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making facilities available for cutting cores from the specimens.

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